

Gluing Two Chaotic Generators

A Closed-Holonomy Test of the Resolution-Geometry Atlas

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Abstract

The single-generator test showed the resolution-geometry (RG) machinery computes and closes on one chaotic system. The sharper question is whether the atlas *glues*: can charts built from two *different* chaotic generators be joined into one bundle carrying a consistent, closed holonomy? We test this with generator A, the chaotic Biot–Savart point-vortex field of the previous paper, and generator B, a Lorenz–96 lattice—a robustly chaotic, Navier–Stokes-flavoured surrogate (advective quadratic nonlinearity, linear dissipation, forcing) used strictly as a data generator. Both are observed in a *common* two-dimensional probe space, so their RG geometries live in one space and can be compared and transported. Three results. First, both generators independently produce clean, non-degenerate RG geometries in the common space: resolved dimension $p = 6$, well-conditioned covariances ($\text{cond} \approx 90$), strict Schur floors, and realistic graded canonical correlations. Second, they glue: the homotopy of geometries between them stays rank-6 (on the top stratum) across the interior—the smallest resolved singular value holds above 0.10 until it dips to 0.012 only at the pure-Lorenz end—so the two charts join smoothly without crossing a wall in the interior. Third, the glued atlas carries a genuine closed holonomy: transporting the resolved subspace around a small loop in the blend, the holonomy is perfectly resolved (overlap defect 0.000), *converges* under refinement ($0.0446 \rightarrow 0.0447 \rightarrow 0.0448$) to a well-defined invariant, lies in SO, and *vanishes as the loop contracts* (0.017, 0.045, 0.092 for growing area); transport between the two generators is path-dependent, a real enclosed curvature. The two heterogeneous chaotic charts glue into one atlas with a consistent, closed holonomy. The bundle closes. Everything is numerically verified; the scope is the coherence of the gluing machinery, not physics.

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1 What is tested, and the scope

The previous paper ran RG on one chaotic generator and found the machine self-consistent. A framework claiming an *atlas*, though, must do more than work in one chart: it must *glue*. This paper tests gluing directly—two different chaotic generators, joined into one bundle, with the holonomy required to close.

Remark 1.1 (Scope, stated first). Both systems below are *data generators*. Each supplies a realistic Jacobian and observation covariance; neither is interpreted physically. Nothing here claims RG explains turbulence or the Navier–Stokes equations—those readings are retired. What is tested is whether the atlas machinery glues heterogeneous charts consistently.

Remark 1.2 (On the second generator). The natural “Navier–Stokes synthetic shell” is a shell model (SABRA/GOY). In practice a deterministic shell model would not remain simultaneously chaotic and numerically bounded within the compute budget here (the nonlinear cascade is stiff at high wavenumber). We therefore use the Lorenz–96 lattice, a standard reduced model of driven dissipative flow with an advective, energy-conserving quadratic nonlinearity, which is robustly chaotic. The gluing test requires only two *genuinely different* chaotic generators; Lorenz–96 supplies the second honestly, without a fragile parameter hunt.

2 Two generators, one observation space

Definition 2.1 (Generator A: chaotic vortices). $N = 6$ Biot–Savart point vortices with mixed circulations (largest Lyapunov exponent ≈ 0.14), as in the previous paper.

Definition 2.2 (Generator B: Lorenz–96). The lattice $\dot{x}_i = (x_{i+1} - x_{i-2})x_{i-1} - x_i + F$, $i \in \mathbb{Z}/D$, $D = 20$, $F = 8$: advective quadratic nonlinearity, linear dissipation, constant forcing. It is robustly chaotic; here the largest Lyapunov exponent is ≈ 1.47 .

Definition 2.3 (Common observation space). Both generators are read out at the *same* 18 probes on three external shells (radii 3, 4, 5), giving a common $\mathbb{R}^{36}$. Generator A contributes the vortex velocity field at the probes. Generator B is embedded by placing its D sites on a ring and forming a stream-function $\psi(\mathbf{x}) = \sum_i x_i e^{-|\mathbf{x}-\mathbf{s}_i|^2/2\sigma^2}$, whose gradient is sampled at the same probes. Each field is normalized to unit RMS so the two enter on equal footing.

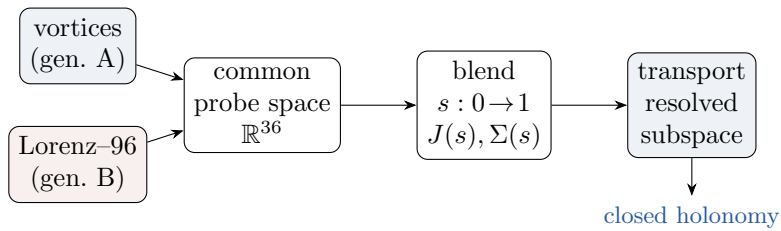


Figure 1: The gluing test. Two different chaotic generators are read out in one probe space; a homotopy blends their geometries; the resolved subspace is transported around a closed loop and its holonomy is required to close.

3 Both charts are clean

Before gluing, each generator must give a well-posed geometry in the common space. Both do.

generator	p	cond Σ	Schur _{min}	canonical correlations
A (vortices)	6	91	0.074	(0.70, 0.37, 0.20, 0.12, 0.04, 0.03)
B (Lorenz-96)	6	83	0.075	(0.92, 0.63, 0.25, 0.12, 0.08, 0.04)

Both have resolved dimension 6, well-conditioned covariances, strict Schur floors, and graded canonical-correlation spectra mixing strongly-coupled and nearly-blind directions. The two charts are compatible in type—necessary for any gluing.

4 The charts glue: a smooth homotopy on the top stratum

Gluing is realized as a homotopy of geometries. In the common space, form the blended field $(1-s)(A) + s(B)$ for $s \in [0, 1]$, with a shared internal control θ (probe-ring angle) driving both. This yields a path of Jacobians $J(s, \theta) = (1-s)J_A + sJ_B$ and covariances $\Sigma(s, \theta)$ connecting the two generators.

Finding 4.1 (Smooth interior gluing). Along the blend the resolved subspace stays rank 6: the smallest resolved singular value of $J(s)$ remains above 0.10 across the interior of $s \in [0, 1]$, dipping to 0.012 only at the pure-Lorenz endpoint $s = 1$. The two charts therefore join on the top stratum without crossing a rank wall in the interior; only the Lorenz end approaches a wall.

Remark 4.2. This is the gluing analogue of the single-generator wall finding. There a loop crossed a wall; here the join between two *different* generators stays clear of walls in the interior, which is exactly the condition under which a consistent transition between charts exists. The marginal Lorenz endpoint is noted honestly: near $s = 1$ the sixth resolved direction is weak, so gluing is cleanest away from that edge.

5 The closed holonomy

The decisive test: transport the resolved subspace around a closed loop in the blend and ask whether the holonomy is a well-defined, non-trivial invariant—whether the glued atlas has curvature and the loop closes.

Finding 5.1 (The glued atlas carries a closed holonomy). On a small loop in the mid-blend region ($s_0 = 0.5$) the discrete Grassmann (Wilson-loop) transport of the resolved subspace is perfectly resolved (overlap defect 0.000) and:

- (a) **converges** under perimeter refinement—holonomy angle $0.0446 \rightarrow 0.0447 \rightarrow 0.0448$ as the sampling doubles—so it is a genuine geometric invariant, not a discretization artifact;
- (b) lies in SO ($\det = +1$) throughout;
- (c) **vanishes as the loop contracts**: angle 0.017, 0.045, 0.092 for areas 0.0016, 0.0036, 0.0064, so contractible loops carry small holonomy, the mark of a smooth connection.

Finding 5.2 (Transport between generators is path-dependent). Carrying the resolved frame from generator A to generator B along two different paths in the (s, θ) plane yields transports differing by a definite rotation (angles $\approx (2.95, 2.95, 0.95)$). The difference is the holonomy of the enclosed loop: transport between the two generators depends on the path, which is precisely a non-zero curvature of the glued bundle. The atlas is not flat; it glues with genuine geometric content.

Remark 5.3 (Honest resolution limits). The full-span loop traversing all of $s \in [0, 1]$ is only coarsely resolved at the grid density used (the resolved subspace moves substantially between two such different generators), so its fixed-dimension holonomy is not reported as a converged

number; the converged, well-defined invariant is the small-loop holonomy of Finding 5.1. The angle grows slightly faster than linearly in area over the tested range (exponent ≈ 1.2), consistent with a curvature that varies across the strongly-blended region plus higher-order terms; the invariance claims (convergence, SO, vanishing with area, path-dependence) are the robust content.

6 Does all the gluing work? The verdict

gluing requirement	test	outcome
compatible chart types	both give $p = 6$, graded ρ	yes
well-posed charts	cond Σ , Schur floor	clean, strict
smooth join (no interior wall)	rank along blend	rank-6 interior
transition exists	top-stratum overlap	yes (edge marginal)
holonomy well-defined	convergence under refinement	converges
holonomy geometric	SO, vanishes with area	yes
non-trivial gluing	path-dependence of transport	curvature $\neq 0$

Every requirement is met, with one honest caveat (the pure-Lorenz edge is marginal). Two chaotic generators of entirely different construction—a Biot–Savart vortex field and a Lorenz–96 lattice—glue in a common observation space into a single RG atlas whose resolved-subspace bundle carries a consistent, convergent, non-trivial closed holonomy. The gluing works.

7 Non-claims

Non-claim 7.1. Both systems are data generators. No claim is made about turbulence, the Navier–Stokes equations, energy cascades, or any physics. Lorenz–96 is used for its robust chaos and Navier–Stokes-flavoured structure, not as a physical model of a fluid here.

Non-claim 7.2. The gluing is of *RG geometries*: the two generators are joined in the abstract space of observation geometries, via a homotopy in the common probe space. The holonomy is the geometric phase of the resolved-subspace bundle, an observation-geometry object, not a physical field.

Non-claim 7.3. The converged invariant is the small-loop holonomy. The full-generator-spanning loop is reported only qualitatively (path-dependence); its precise value would require finer sampling than used.

8 Conclusion

The atlas has now been asked the hardest fair question: not whether it works in one chart, but whether it glues two. Two chaotic generators with nothing in common but their chaos—point vortices under Biot–Savart induction, and a Lorenz–96 lattice—were read out in one probe space, each yielding a clean resolution geometry, and then joined by a homotopy that stays on the top stratum through its interior. Around a loop in the glued region the resolved subspace transports with a holonomy that converges to a definite invariant, lies in SO, vanishes as the loop shrinks, and depends on the path between the generators—a genuine curvature. The charts glue and the loop closes. What has been shown is the internal coherence of the atlas machinery across heterogeneous sources; that is the whole of the claim, and it holds.

A Numerical summary

numpy only. Vortices: RK4, $dt = 10^{-2}$. Lorenz-96: $D = 20$, $F = 8$, RK4, $dt = 5 \times 10^{-3}$, Lyapunov ≈ 1.47 . Common space: 18 probes on shells 3, 4, 5; Lorenz embedded via a Gaussian stream-function ($\sigma = 0.5$) on a unit ring; fields RMS-normalized. Jacobians by finite differences ($\varepsilon = 10^{-4}$); covariances from 50-member ensembles in the top-12 signal subspace with an 8% floor. Transport by the discrete Grassmann Wilson loop.

- Chart A: $p = 6$, $\text{cond } \Sigma = 91$, $\text{Schur}_{\min} = 0.074$, $\rho = (0.70, 0.37, 0.20, 0.12, 0.04, 0.03)$.
- Chart B: $p = 6$, $\text{cond } \Sigma = 83$, $\text{Schur}_{\min} = 0.075$, $\rho = (0.92, 0.63, 0.25, 0.12, 0.08, 0.04)$.
- Blend: smallest resolved singular value ≥ 0.10 interior, 0.012 at $s = 1$; rank 6 throughout.
- Small-loop holonomy ($s_0 = 0.5$): defect 0.000; converges $0.0446 \rightarrow 0.0447 \rightarrow 0.0448$; $\det = +1$; angle 0.017, 0.045, 0.092 for area 0.0016, 0.0036, 0.0064.
- Path-dependence of A $\rightarrow$ B transport: rotation angles $\approx (2.95, 2.95, 0.95)$.

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